

(a) physical coordinates

mate by closest-point search

body A

$$\mathbf{t} = t_N \mathbf{n} + \mathbf{t}_T$$

$$\mathbf{x} = \varphi_A(\theta_A)$$

closest-point
search (iterative)

$$\varphi_B(\psi_B)$$

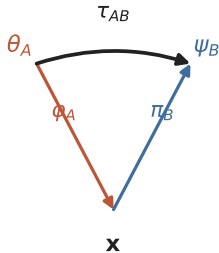
body B

$\partial\Omega_A$

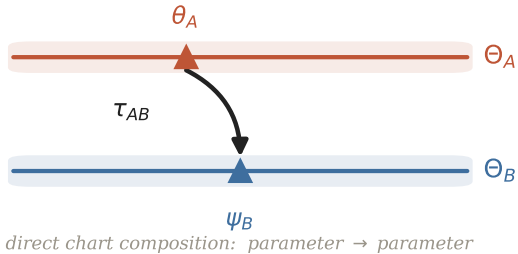
$\partial\Omega_B$

$$\tau_{AB} = \pi_B \circ \varphi_A$$

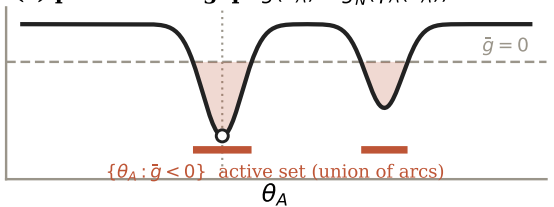
evaluate φ_A , invert φ_B : no search



(b) reference coordinates



(c) pulled-back gap $\bar{g}(\theta_A) = g_N^B(\varphi_A(\theta_A))$



$$\delta W_c = \int_{\Gamma_c} (t_N \delta g_N + \mathbf{t}_T \cdot \delta \mathbf{g}_T) dA \quad \text{depends on the geometry only through } (g_N, \mathbf{n})$$